

CodeGraph — Architecture Overview

Git blame tells you who changed it. CodeGraph tells you why.

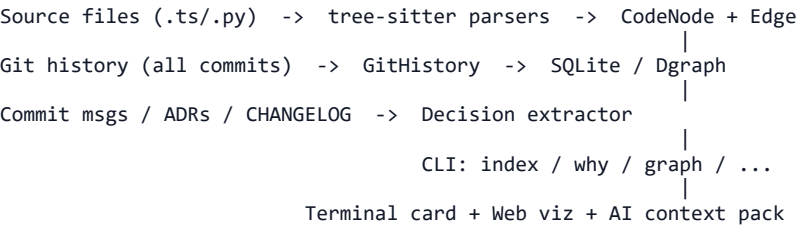
Problem

Developers constantly ask "why is this code written this way?" git blame shows authorship. GitHub search shows text matches. ADR tools require manual writing. AI coding assistants see only the current file. No existing tool connects a line of code to the design decisions that shaped it.

Solution — three layers, one graph

- 1. Structure: tree-sitter parses TypeScript/Python into AST nodes and edges (uses / defined_by / contains)
- 2. Time: Every node carries valid_from / valid_to, derived from real Git history
- 3. Decisions: Extracted from commit messages, ADRs, and CHANGELOG, linked to the exact code lines they justify

Architecture



Data model

CodeNode: uid, kind, name, file_path, line_start, line_end, language, valid_from, valid_to, parent_uid
Edge: from_uid, to_uid, kind (uses / defined_by / contains)
Decision: content, reason, alternatives, status, source, source_ref, timestamp, author, constraints, confidence

Seven commands

codegraph index <path>	Parse + ingest Git history + store
codegraph why <file>:<line>	Decision card for that line
codegraph graph --at <date>	Code graph at a past moment
codegraph decisions --timeline	Decision evolution chain
codegraph conflicts	Detect contradictory decisions
codegraph export --for-ai	Export AI agent context
codegraph serve	Local web visualization

Tech choices

tree-sitter	Multi-language parsing, incremental, industry standard
GitPython	Read-only access to real Git history
Dgraph	Native graph queries for multi-hop decision tracing
Typer + Rich	Fast CLI, readable terminal output
pydantic	Typed data models
SQLite	Default backend; zero-config, local-first

Status (as of submission)

Done	Product site (HTML/CSS/JS)
Done	CLI with 6 commands
Done	tree-sitter parsing for TypeScript and Python
Done	SQLite storage
Done	Real Git history traversal
Done	Decision extraction from commits, ADRs, CHANGELOG
Done	Real time-travel queries (graph --at)
Done	25 tests passing
In progress	Dgraph backend
Planned	Web visualization (Cytoscape.js)
Planned	VSCode extension

Roadmap

v0.1.0	CLI + SQLite (current)
v0.2.0	Real Git history + decision extraction
v0.3.0	Dgraph backend + time-travel queries
v0.4.0	Web visualization
v0.5.0	LLM-based decision extraction (local-first, Ollama)
v0.6.0	VSCode extension

Built by a 15-year-old self-taught developer. · MIT License. · <https://github.com/fourwich/codegraph>